

Assignment: Natural Language Processing (NLP) for Text Classification in Python

Objective

The objective of this assignment is to learn the basic workflow of text classification using Natural Language Processing (NLP) techniques in Python. You will preprocess text data, extract features, build a machine learning model, and evaluate its performance.

```
In [10]: import warnings
import sys
if not sys.warnoptions:
    warnings.simplefilter("ignore")
```

Task 1: Import Necessary Libraries

```
In [12]: from sklearn.feature_extraction.text import CountVectorizer
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score, classification_report
```

Task 2: Create a Dataset

```
In [14]: # Sample dataset
texts = [
    "I love this product",
    "This is terrible",
    "Absolutely fantastic!",
    "Not good at all",
    "Worth every penny",
    "Awful experience",
    "Excellent quality",
    "I hate it",
    "Superb performance",
```

```
    "Horrible design"  
]  
labels = ["positive", "negative", "positive", "negative", "positive", "negative", "positive", "negative", "positive"]
```

Task 3: Preprocess the Data

```
In [16]: # Preprocessing and Feature Extraction  
vectorizer = CountVectorizer() # Initialize the vectorizer  
X = vectorizer.fit_transform(texts) # Transform text to BoW representation
```

Task 4: Split the Data

```
In [18]: # Split data into training and testing sets  
X_train, X_test, y_train, y_test = train_test_split(X, labels, test_size=0.2, random_state=42)
```

Task 5: Train a Machine Learning Model

```
In [20]: # Train a Naive Bayes classifier  
model = MultinomialNB()  
model.fit(X_train, y_train)
```

```
Out[20]:  
▼ MultinomialNB ⓘ ?  
MultinomialNB()
```

Task 6: Make Predictions

```
In [22]: # Make predictions  
y_pred = model.predict(X_test)
```

Task 7: Evaluate the Model

```
In [24]: # Evaluate model performance  
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
print("Classification Report:\n", classification_report(y_test, y_pred))
```

Accuracy: 0.0

Classification Report:

	precision	recall	f1-score	support
negative	0.00	0.00	0.00	1.0
positive	0.00	0.00	0.00	1.0
accuracy			0.00	2.0
macro avg	0.00	0.00	0.00	2.0
weighted avg	0.00	0.00	0.00	2.0